ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 1 · PRACTICE D

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET first paper. Section A (30): multiple choice, true or false, matching and acronyms · Section B (70): types and uses of robots, 3D printing, ultrasonic distance, degrees of freedom, batteries and power calculations, a night-patrol robot scenario and a servo-arm program to study.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL calculations and write units with the answers.
6. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Which component changes electrical energy into rotary motion?

A LED
B Resistor
C DC motor
D Capacitor

1.2 The part of a mobile robot that runs its program and makes decisions is the …

A microcontroller.
B wheel.
C chassis.
D battery.

1.3 Which sensor measures tilt and acceleration?

A LDR
B Thermistor
C Microphone
D Accelerometer

1.4 A robot arm has five rotary joints. How many degrees of freedom does it have?

A 3
B 5
C 6
D 1

1.5 The Arduino function that runs over and over again is …

A loop().
B setup().
C pinMode().
D begin().

1.6 Which module is commonly used to control a robot from a cellphone over a short distance?

A GPS module
B LCD module
C HC-05 Bluetooth module
D Relay module

1.7 An end effector that uses suction to pick up flat objects is a …

A magnetic gripper.
B vacuum gripper.
C welding torch.
D spray gun.

1.8 A potentiometer used as a sensor in a robot joint measures …

A light.
B sound.
C humidity.
D angle (position).

1.9 The voltage on an Arduino Uno digital pin set HIGH is about …

A 5 V.
B 12 V.
C 0 V.
D 230 V.

1.10 Which type of robot moves by walking?

A Gantry robot
B Legged robot
C SCARA robot
D Conveyor robot

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.5) in the ANSWER BOOK.

2.1 A hobby servo motor normally has three wires: power, ground and signal.

2.2 Sound travels faster than light.

2.3 PWM can be used to dim an LED.

2.4 analogRead() on an Arduino Uno gives a value from 0 to 255.

2.5 Lithium batteries must be charged with the correct charger to prevent overheating and fire.

(5)

[5]

QUESTION 3

Choose a description from COLUMN B that matches an item in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.10) in the ANSWER BOOK.

	COLUMN A		COLUMN B
3.1	Chassis	A	Keeps the supply voltage constant
3.2	Wheel encoder	B	Holds and releases objects
3.3	Gripper	C	Frame that holds all the parts of the robot
3.4	Accelerometer	D	Reduces speed and increases torque
3.5	Voltage regulator	E	Wireless link to a cellphone or tablet
3.6	Bluetooth module	F	Measures how far a wheel has turned
3.7	Limit switch	G	Detects that a moving part has reached the end of its travel
3.8	Servo horn	H	Measures acceleration and tilt
3.9	Arduino IDE	I	Software used to write and upload sketches
3.10	Gearbox	J	Small arm fixed to the shaft of a servo

(10)

[10]

QUESTION 4

Write out the full form of each acronym below.

4.1 DOF

4.2 LED

4.3 IoT

4.4 USB

4.5 AI

(5)

[5]

TOTAL SECTION A: [30]

SECTION B

QUESTION 5

5.1 Explain the difference between an autonomous robot and a remote-controlled robot, and give ONE example of each. (4)

5.2 Explain how 3D printing is used to make parts for robots, and give TWO advantages of using it. (4)

5.3 Explain how an ultrasonic sensor works out the distance to an object. (4)

5.4 Explain what is meant by degrees of freedom, using a robot arm as an example. (4)

5.5 Give FOUR ways in which robots are used in South African industries. (4)

[20]

QUESTION 6

6.1 Explain the difference between connecting two robot batteries in series and connecting them in parallel. (4)

6.2 Two 3,7 V, 2 000 mAh lithium cells are connected in series. Write down the total voltage and the capacity of the pack. (2)

6.3 A robot motor draws 0,8 A from a 6 V supply.

6.3.1 Calculate the power used by the motor. (2)

6.3.2 Calculate the resistance of the motor. (2)

6.4 An ultrasonic sensor measures an echo time of 2 900 µs. The speed of sound is 343 m/s. Calculate the distance to the object in centimetres. (4)

6.5 Give TWO factors to consider when choosing a battery for a mobile robot. (2)

[16]

QUESTION 7

A school wants a small robot that patrols the corridors at night. It drives on two wheels, uses a PIR sensor to detect people, a camera to take photos, and a buzzer as an alarm. An ESP32 board controls the robot and sends a message to the security guard's cellphone over Wi-Fi when someone is detected.

7.1 How does the PIR sensor detect a person? (2)

7.2 Why is an ESP32 used here instead of an Arduino Uno? (2)

7.3 Why does the camera need infrared (IR) LEDs to take useful photos at night? (2)

7.4 Give TWO things to consider about powering the robot for a whole night. (2)

7.5 Give ONE privacy or ethical concern about a robot that takes photos of people, and how it can be handled. (2)

[10]

QUESTION 8

A student has built a small robot arm with two servo motors (base and arm). Two potentiometers act as "joysticks" to move the arm, and a push button locks the arm in position. An LED shows when the arm is locked.

Study the code below and answer the questions that follow.

```
 1  #include <Servo.h>
 2  Servo baseServo;
 3  Servo armServo;
 4  const int basePot = A0;
 5  const int armPot = A1;
 6  const int buttonPin = 2;
 7  const int ledPin = 13;
 8  bool locked = false;
 9
10  void setup() {
11    baseServo.attach(9);
12    armServo.attach(10);
13    pinMode(buttonPin, INPUT_PULLUP);
14    pinMode(ledPin, OUTPUT);
15  }
16
17  void loop() {
18    if (digitalRead(buttonPin) == LOW) {     // button pressed
19      locked = !locked;                      // lock or unlock the arm
20      digitalWrite(ledPin, locked);
21      delay(300);
22    }
23    if (!locked) {
24      int baseAngle = map(analogRead(basePot), 0, 1023, 0, 180);
25      int armAngle = map(analogRead(armPot), 0, 1023, 20, 160);
26      baseServo.write(baseAngle);
27      armServo.write(armAngle);
28    }
29    delay(15);
30  }
```

8.1 What is the purpose of line 1 and of lines 2 and 3? (4)

8.2 Explain what the map() function does in line 24, and calculate the base angle when analogRead(basePot) gives 512. (4)

8.3 Why does line 25 limit the arm to angles from 20° to 160°? (4)

8.4 Explain what happens each time the button is pressed (lines 18–22), including what line 19 does. (4)

8.5 Why is there a delay of 300 ms in line 21? (4)

8.6 What does INPUT_PULLUP do in line 13, and why does the button read LOW when it is pressed? (4)

[24]

TOTAL SECTION B: [70]

TOTAL: 100

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END OF QUESTION PAPER